Hamoud Aljamaan



Contact

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Profiles

Education

Doctor of Philosophy
Computer Science, 2015
University of Ottawa, Canada

Master of Science
Computer Science, 2009
KFUPM, Saudi Arabia

Bachelor of Science
Computer Science, 2006
King Saud University, Saudi Arabia

Technical Skills

Python, Java, C/C++
PyTorch, TensorFlow/Keras
scikit-learn, pandas, NumPy
Git, UML, LaTeX

Languages

Arabic: native
English: fluent

Academic Appointments

King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia
Information and Computer Science Department

Associate Professor Oct 2025–present
Assistant Professor Jan 2016–Oct 2025
Chairman Jan 2019–Aug 2023

Teaching: software engineering and artificial intelligence.

Research Interests

- AI for Software Engineering
- Ensemble Learning
- Software Quality
- Software Security

Selected Leadership Achievements

Programs: Launched AI/ML and Cybersecurity/Blockchains concentrations; AI and Cybersecurity master's programs; women's undergraduate and graduate programs.
Industry: Established Aramco 4IR Summer School (since 2019) and the joint Cybersecurity Chair.

Quality: Led CS/SWE curriculum revisions, ABET re-accreditation (2021), and NCA alignment (2022).

Recruitment: 8 graduate assistants and 14 faculty members.

Selected Publications and Patent

24 journal articles, 17 conference papers, and one granted U.S. patent.

- S. J. Alsunaidi, **H. Aljamaan**, M. Hammoudeh. "Leveraging Machine Learning Models to Improve Smart Contract Security: A Survey of Vulnerabilities and Detection Methods." *ACM Computing Surveys*, 58(6), Article 151 (2026).
- S. J. Alsunaidi, **H. Aljamaan**, M. Hammoudeh. "DIVE: A Multi-Label Smart Contract Vulnerability Dataset." *Scientific Data*, 13, Article 664 (2026).
- A. Alazba, **H. Aljamaan**, M. Alshayeb. "CoRT: Transformer-based code representations with self-supervision by predicting reserved words for code smell detection." *Empirical Software Engineering*, 29(3), Article 59 (2024).
- A. Alazba, **H. Aljamaan**. "Code smell detection using feature selection and stacking ensemble: An empirical investigation." *Information and Software Technology*, 138, Article 106648 (2021).

Granted U.S. patent: US 12,591,499 B2, code smell detection using self-supervised transformer representations; granted 31 March 2026.

Selected Funding and Supervision

- **Cost Center Manager and Co-I, ICS-2506:** Aramco 4IR Summer School project; approximately SAR 2.5M (2020–2024).

Thesis supervision: 2 PhD and 7 MSc theses completed.

Selected Awards and Professional Service

- **2025 awards:** Research Excellence Award for Journal Publications (IRC-FDE) and Distinction Award for Excellence in Academic Advising (KFUPM).
- **ASSURE-AI 2027 (upcoming):** Organizing Co-chair and Proceedings Chair.